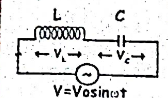
3) L-C CIRCUIT

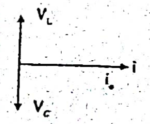


$V = V_0 \sin\omega t$

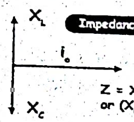
$V_L = i_0X_L\ V_C = i_0X_C$

Voltage phasor diagram

$V = V_L \sim V_C$ [ie, $(V_L - V_C)$ or $(V_C - V_L)$]



Impedance Phasor Diagram



$Z = X_L \sim X_C$ [$(X_L - X_C)$ or $(X_C - X_L)$]

if $X_L > X_C$, Voltage leads the current by $\frac{\pi}{2}$

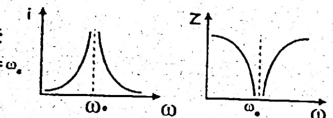
if $X_C > X_L$, current leads the voltage by $\frac{\pi}{2}$

if $X_L = X_C$, Z = 0, i = ∞

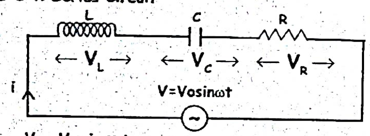
or,

$\omega L = \frac{1}{\omega C}$

$\omega = \frac{1}{\sqrt{LC}} = \omega_0$



L-C-R Series Circuit

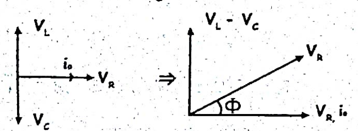


$V = V_0 \sin\omega t$

$V_R = i_0R\ V_L = i_0X_L\ V_C = i_0X_C$

Assuming $V_L > V_C$ for drawing phasor

Voltage phasor diagram

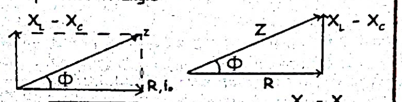


$V = \sqrt{V_R^2 + (V_L - V_C)^2}$

Here $i = i_0 \sin(\omega t - \Phi)$

(since V_L is leading)

Impedance Triangle



$Z = \sqrt{R^2 + (X_L - X_C)^2}$ $\tan\Phi = \frac{X_L - X_C}{R}$

RESONANCE IN LCR SERIES CIRCUIT

In series resonance, impedance of circuit is minimum & equal to resistance ⇒Z= R, and curent is maximum

Condition for resonance

$X_L = X_C \Rightarrow L\omega = \frac{1}{C\omega}$

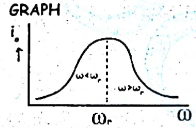
$\omega = \omega_r = \frac{1}{\sqrt{LC}}$ rad / sec

ω_r → resonant frequency (angular)

$f = f_r = \frac{1}{2\pi\sqrt{LC}}$ H_z

f_r = resonant frequency

GRAPH



Variation of peak current with applied frequency

$\omega < \omega_r$, $X_C > X_L$ current leads

$\omega > \omega_r$, $X_L > X_C$ current lags

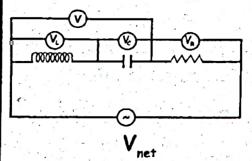
In resonance

$V = V_R$ (applied voltage = voltage across resistance)

Z = R (impedance is minimum and equal to resistance)

Voltmeter connected across V_L & V_C will show the same reading

Voltmeter connected commonly across inductor & capacitor shows no reading



Here V = 0

$V_L = V_C$

$V_N = V_R$

$i_{max} = \frac{V_{net}}{Z} = \frac{V_R}{R}$

APPLICATION OF RESONANT CIRCUIT

Tuning mechanism of a radio or TV set

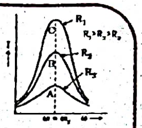
1. Antenna of radio accepts signals
2. Signal acts as an AC source in tuning the radio
3. In tuning, capacitance of capacitor is varied such that the resonant frequency of the circuit becomes nearly equal to the frequency of the radio signal received.

So, the simple is largely amplified and distinctly heard

QUALITY FACTOR

$Q = \frac{\omega_r L}{R} = \frac{1}{\omega_r CR} = \frac{1}{R}\sqrt{\frac{L}{C}}$

or $Q = \left[\frac{\text{Voltage across C or L}}{\text{applied voltage}}\right]_{resonance}$



Less sharp the resonance, less is the selectivity of the circuit. If the Quality factor is large, R is low or L is large, the circuit is more selective.

Sharpness of Resonance

Sharpness= $Q = \frac{\omega_r}{2\Delta\omega}$; 2Δω -bandwidth

smallerΔω, sharper or narrower the resonance.

POWER IN AC CIRCUIT

Average Power $\overline{P} = V_{rms} I_{rms} \cos\Phi$

$\overline{P} = I_{rms}^2 Z \cos\Phi$

Case 1

Purely Resistive circuit - Φ=0,   COS Φ =1

Maximum power dissipation

Case 2

Purely inductive or capacitive circuit-

Φ =90°   cosΦ=0

No power is dissipated even though a current is flowing in the circuit

Case 3

LCR Series circuit

Φ non zero in R-L, C-R, or CLR circuit.

$\overline{P} = V_{rms} I_{rms} \cos\Phi$

Case 4

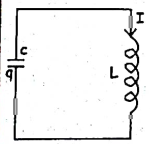
Power dissipation at resonance

$X_L - X_C = 0$ or Φ=0   ⇒ cosΦ=1   ⇒Z=R

$P = I^2Z = I^2R$

Maximum power is dissipated in a circuit at resonance.

LC OSCILLATIONS



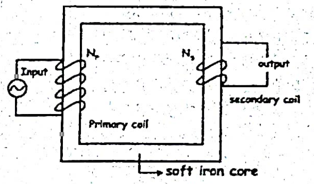
Capacitor ⟶ stores electrical energy

Inductor ⟶ stores magnetic energy

When connected, charge on the capacitor and current in the inductor perform electrical oscillations between each other.

TRANSFORMERS

"Device which raises or lowers voltage in ac circuits through mutual induction". Transformer can increase or decrease voltage or current but not both simultaneously.



EQUATIONS

1) $\frac{V_S}{V_P} = \frac{N_S}{N_P} = \frac{I_P}{I_S}$

2) Efficiency $\eta = \frac{P_{out}}{P_{in}} = \frac{V_S I_S}{V_P I_P}$

3) For ideal transformer, η=1

V_s -Voltage in secondary

V_p -Voltage in primary

N_s -No of turns in secondary

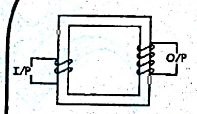
N_p -No of turns in primary

I_p -Current in primary

I_s -Current in secondary

TRANSFORMER TYPES

step up transformer



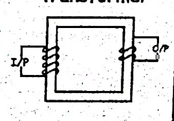
$N_P < N_S$

$V_P < V_S$

$I_P > I_S$

$R_P < R_S$

step down transformer



$N_P > N_S$

$V_P > V_S$

$I_P < I_S$

$R_P > R_S$

LOSSES IN TRANSFORMER

1) Cu loss (I^2R loss)
   → To minimise, windings are made of thick Cu wires (high resistance)
2) Eddy current loss
   → To minimise Cores are laminated
3) Hysteresis loss
   → select material of narrow hysteresis loop
   → Cores of transformer is made of soft iron
4) Magnetic flux linkage
   → To minimise, secondary winding is kept inside the primary winding
5) Humming loss

EM WAVES

ELECTRO MAGNETIC WAVES

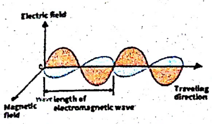
- Time varying electric and magnetic fields that propogate in space
- Oscillating electric and magnetic fields are mutually perpendicular to each other and both are perpendicular to direction of propagation
- Speed of wave = Speed of light = 3×10^8 m/s

SOURCE OF ELECTROMAGNETIC WAVES IS ACCELERATING / OSCILLATING CHARGE

ENERGY DENSITY

1. U_E (Electric field) $= \frac{1}{2}\varepsilon_0 E^2$
2. U_B (Magnetic field) $= \frac{1}{2}\frac{B^2}{\mu_0}$
3. Average energy density $\bar{u}_E = \frac{1}{4}\varepsilon_0 E^2$, $\bar{u}_B = \frac{1}{4}\frac{B^2}{\mu_0}$
4. $\bar{u}_E = \bar{u}_B$
5. Total average energy density
   $u = u_E + u_B = \frac{1}{2}\varepsilon_0 E^2 = \frac{1}{2}\frac{B^2}{\mu_0}$ $[u_E = u_B]$

TRANSVERSE NATURE OF EM WAVES



$Ey = E_0 \sin(\omega t - kx)$

$Bz = B_0 \sin(\omega t - kx)$

$E_0 = C B_0$

INTENSITY OF WAVE

Energy crossing per unit time area perpendicular to the direction of wave propagation

$$\text{Intensity} = \frac{\text{Energy}}{\text{time} \times \text{area}} = \frac{\text{Power}}{\text{area}}$$

FORMULAE TO REMEMBER

$$I = \frac{1}{2}\varepsilon_0 E^2 \times c$$

$$I = \frac{B^2}{2\mu_0} \times c$$

SPEED OF EM WAVE (VACUUM) $c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} = 3 \times 10^8$ m/s

SPEED OF EM WAVE (MEDIUM) $V_{med} = \frac{1}{\sqrt{\mu_0 \mu_r \varepsilon_0 \varepsilon_r}} = \frac{1}{\sqrt{\mu_m \varepsilon_m}}$

REFRACTIVE INDEX $\mu_{med} = \frac{c}{V_{med}} = \sqrt{\mu_r \varepsilon_r}$

REMARKS

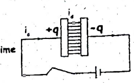
$$C = \frac{E_0}{B_0}, \quad C = \frac{\omega}{K}$$

Maximum electric force $F_{E(max)} = qE_0$

Maximum magnetic force $F_{B(max)} = qvB_0$

DISPLACEMENT CURRENT

Displacement current - Current in vacuum or dielectric when electric field is changing with time



$I_d = \varepsilon_0 \frac{d\phi_E}{dt}$

Displacement current = Conduction current

POYNTING VECTOR

$$S = \frac{\text{Energy}}{\text{time} \times \text{area}} = \frac{\text{Power}}{\text{area}} \qquad \vec{S} = \frac{\vec{E} \times \vec{B}}{\mu_0}$$

POYNTING VECTOR

- Magnitude represents power per unit area
- Direction is along the direction of wave propagation
- SI unit :- $\frac{\text{Joule}}{\text{sec } m^2}$, $\frac{\text{Watt}}{m^2}$

MOMENTUM OF EM WAVES

1) $P = \frac{U}{c}$ (If wave is completely absorbed)
2) $P = \frac{2U}{c}$ (If wave is completely reflected)

MAXWELL'S EQUATIONS

$\oint \vec{E}.d\vec{A} = \frac{Q}{\varepsilon_0}$ [Gauss's Law of Electrostatics]

$\oint \vec{B}.d\vec{A} = 0$ [Gauss's Law of Magnetism]

$\oint \vec{E}.d\vec{l} = \frac{-d\phi_B}{dt}$ [Faraday's Law of Electromagnetic Induction]

$\oint \vec{B}.d\vec{l} = \mu_0 (i_c + i_d) = \mu_0 i_c + \mu_0 \varepsilon_0 \frac{d\phi_E}{dt}$ [Ampere-Maxwell's Law]

ELECTROMAGNETIC SPECTRUM

Radio Waves
Produced by : Accelerated motion of charges in conducting wires
Frequency : 500 kHz - 1000MHz
Application : Cellular phones

Microwaves
Produced by : Special vacuum tubes - Klystrons, magnetrons, Gunn diodes
Detection by : Point contact diodes
Application : Radar systems, microwave oven in domestic purposes

Infrared waves
Produced by : Vibration of atoms and molecules
Detection by : Thermopiles, Bolometer, Infrared photographic film
Application : Used in remote switches for TV set, maintains average temperature through green house effect, Infrared lamps, Infrared detectors

Visible light
Wavelength : 400nm to 700nm
Frequency : 4×10^{14} Hz to 7×10^{14}

Ultraviolet rays
Wavelength : 4×10^{-7} m to 6×10^{-10} m
Produced by : Very hot bodies
Important source : Sun
Application : LASIK Surgery, UV lamps - Kills germs in purifiers
Detection by : Photocells, photographic film

X-RAYS
Produced by : High energy electrons striking metal targets
Wavelength range : 10nm to 10^{-4}nm
Application : Diagnostic tool, treatment of cancer
Detection : Photographic film, Geiger tubes, Ionisation chamber

GAMMA rays
Produced in : Nuclear reactions, Radioactive decay of nucleus
Wavelength range : 10nm to 10^{-14} nm
Application : In medicine to kill cancer cells.

Decreasing order of wavelength $\longrightarrow$

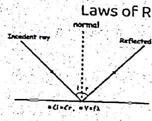
R M I V U X G

$\longrightarrow$ Increasing order of frequency

RAY OPTICS-1

Ray Optics
- Reflection
  - Plane mirror
  - Spherical mirror
- Refraction
  - Plane surface
  - Spherical surface
  - Prism
  - TIR

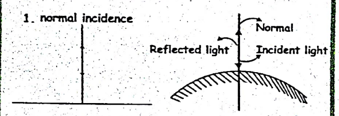
Laws of Reflection



- Angle of incidence=Angle of reflection
- After reflection velocity & frequency of light remains same, but intensity decreases
- phase change of π occurs if light is incident from rare to denser medium

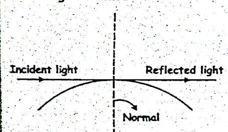
Special Cases:

1. normal incidence



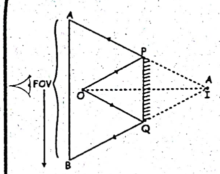
$\angle i = \angle r = 0$

2. Grazing incedence:-



In case light strikes the reflecting surface tangentially $\angle i = \angle r = 90^\circ$

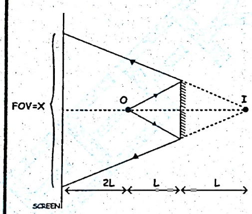
Field of View



Field of view

Field of view defines the area visible from the perspective of observer through mirror.

Finding Field of view



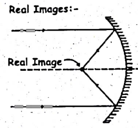
if the distance from observer to mirror is L, then field of view is x
By similar Δs,

$$\frac{x}{4L} = \frac{2L}{L}$$

$x = 4 \times 2L$
$= 8L$

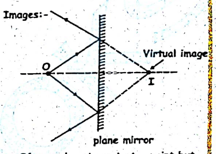
Images & Objects

Real Images:-



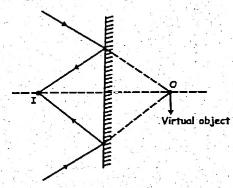
Reflected or refracted rays actually meet or converge at a point

Virtual Images:-



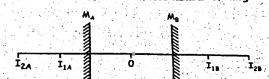
If rays do not meet at a point but appear to meet virtual image is formed

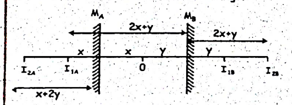
- Reflected Ray(Image):-
  - Diverging-Virtual Image
  - Converging-Real Image
- Incident Ray(Object):-
  - Diverging-Virtual object (see diagram)



Number of Images

If two plane mirrors were placed opposite to each other, there will be infinite number of images

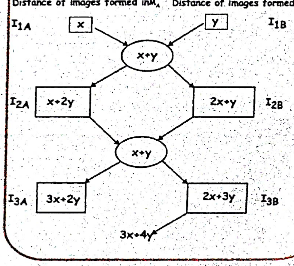




x,y is the distance of object from mirrors M_A&M_B
O is the object, I_{1A}, I_{2B} images

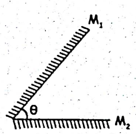
Shortcut to find the distance to images

Distance of images formed in M_A — Distance of images formed in M_B



Number of Images for Inclined Mirror

Let the mirrors be at an angle θ



Method to find number of images

$$\frac{360}{\theta} = n$$

- n=even
  - No.of images=n-1
- n=odd
  - Object lying symmetrically between mirrors → No.of images=(n-1)
  - Object lying unsymmetrically between mirrors → No.of images=(n)

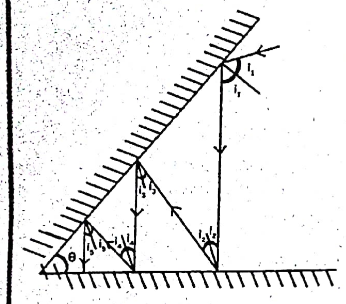
RAY OPTICS-2

Deviation of Rays

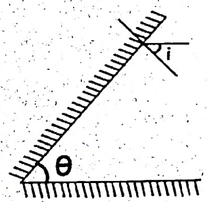
Deviation in single mirror:

Deviation, $\delta = 180 - 2i$

Deviation for two mirrors inclined at an angle



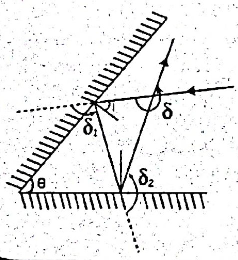
$i_2 = i_1 - \theta$
$i_3 = i_2 - \theta$
$i_4 = i_3 - \theta$
$i_5 = i_4 - \theta$



No.of reflections at which light becomes normal $= \frac{i}{\theta} + 1$

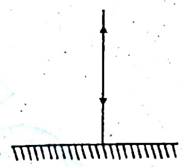
Total no.of reflections $= \frac{i}{\theta} + 1 + \frac{i}{\theta} = \frac{2i}{\theta} + 1$

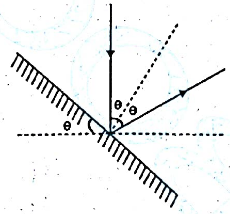
Deviation after two sucessive reflections



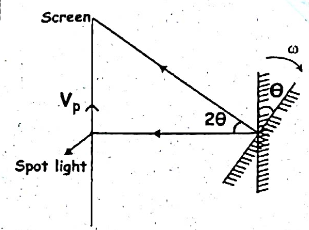
$\theta > i$
$\delta = \delta_1 + \delta_2$
$= 360 - 2\theta$

Effect of rotation of mirror

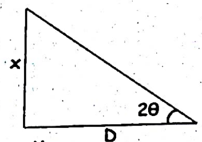




Angle of rotation $= \theta$



Deviation of reflected ray $= 2\theta$



$\tan 2\theta = \frac{x}{D}$

small angle
$\tan 2\theta \approx 2\theta = \frac{x}{D}$

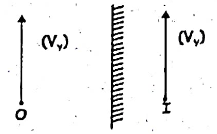
$\Rightarrow \theta = \frac{x}{2D}$

Differentiate, $\Rightarrow \omega = \frac{V_p}{2D}$

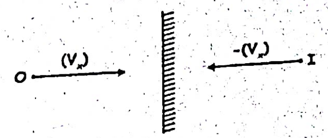
$\Rightarrow V_p = 2\omega D$

Relative motion in plane mirror

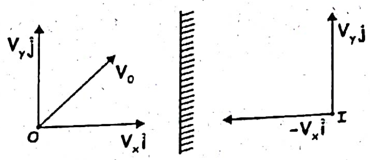
parallel direction



Perpendicular direction



Both parallel and perpendicular:-

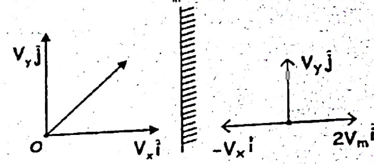


$\vec{V}_o = V_x\hat{i} + V_y\hat{j}$

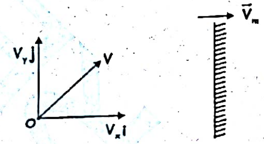
$\vec{V}_I = -V_x\hat{i} + V_y\hat{j}$

Both object and mirror are moving:-

$\vec{V}_m = V_m\hat{i}$



Relative velocity of object with respect to mirror



$V_{o/m} = (V_x - V_m)\hat{i} + V_y\hat{j}$

Relative velocity of image with respect to mirror

$V_{I/m} = -(V_x - V_m)\hat{i} + V_y\hat{j}$

Velocity of image

$V_I = \vec{V}_{I/m} + \vec{V}_m \quad = (2V_m - V_x)\hat{i} + V_y\hat{j}$

Relative velocity of image with respect to object

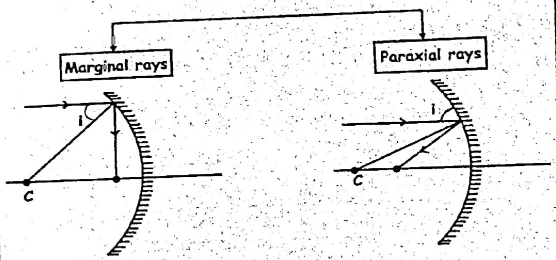
$V_{I/o} = \vec{V}_I + \vec{V}_o$

Spherical mirror

The relationship between angle of incidence and focal length

$f = R - \frac{R}{2\cos i} \quad \Rightarrow f \approx \frac{R}{2}$ (Paraxial rays)

Spherical mirror



To avoid spherical aberration

1) Use small aperture mirror → Avoid marginal
→ Only paraxial

2) Blackening of central portion → Avoid paraxial
→ Only marginal

Magnification and mirror formula

Sign convention and different terminology

1) Radius of curvature (R) :

Distance between pole and center of curvature

2) Focal length (f):

Image point on the principle axis for an object at ∞

Convex $\rightarrow$ +ve

Concave $\rightarrow$ -ve

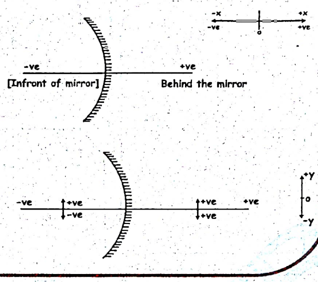
Plane mirror $\rightarrow \infty$

3) Aperture:

Effective diameter of portion of mirror reflecting the light.

Reflecting area α (aperture)^2

Sign convention



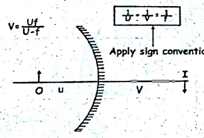
RAY OPTICS-3

Mirror formulae

$V=\frac{Uf}{U-f}$

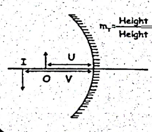
$\boxed{\frac{1}{u}+\frac{1}{v}=\frac{1}{f}}$

Apply sign convention



Transverse magnification

$m_T=\frac{\text{Height of the image}}{\text{Height of the object}}=\frac{h_i}{h_o}$



$\boxed{m_T=-\frac{v}{u}}$

$\boxed{m_T=\frac{f}{f-u}}$

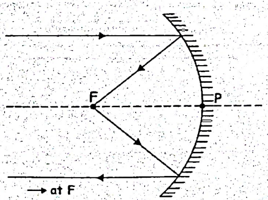
$\boxed{m_T=\frac{f-V}{f}}$

m_T
- $|m_T|>1 \longrightarrow$ magnified
- $|m_T|=1 \longrightarrow$ same size
- $|m_T|<1 \longrightarrow$ small

m_T
- +ve $\Longrightarrow$ Erect
- -ve $\Longrightarrow$ Inverted

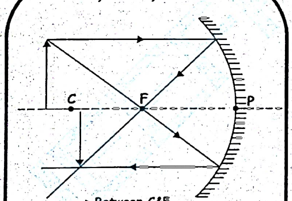
- u, v => Same sign => virtual image
- u, v => Opposite sign => real image

Object at ∞



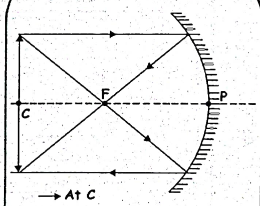
- $\rightarrow$ at F
- $\rightarrow$ Inverted
- $\rightarrow$ Extremely small
- $\rightarrow$ $m<<-1$

Object beyond C



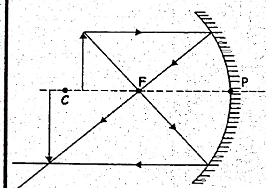
- $\rightarrow$ Between C&F
- $\rightarrow$ Inverted
- $\rightarrow$ small
- $\rightarrow$ $m<-1$

At C



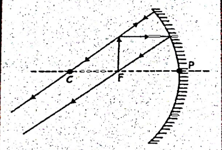
- $\rightarrow$ At C
- $\rightarrow$ Inverted
- $\rightarrow$ same size
- $\rightarrow$ $m=-1$

Between F and C



- $\rightarrow$ Beyond C
- $\rightarrow$ Inverted
- $\rightarrow$ $m>-1$
- $\rightarrow$ Large

At F



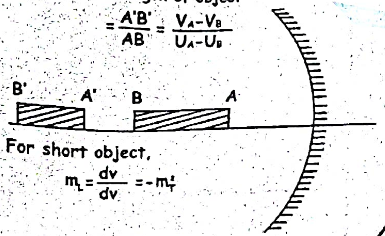
- $\rightarrow$ At ∞
- $\rightarrow$ Extremely large
- $\rightarrow$ $m>>-1$

RAY OPTICS - 4

longitudinal magnification

$m_L = \dfrac{\text{length of image}}{\text{length of object}}$

$= \dfrac{A'B'}{AB} = \dfrac{V_A - V_B}{U_A - U_B}$



For short object,

$m_L = \dfrac{dv}{dv} = -m_T^2$

Relative motion in spherical mirror

Relative velocity of image with respect to spherical mirror

$(V_{I/m}) = -m^2(V_{O/m})$

Relative velocity of object with respect to spherical mirror

$V_{O/m} = \vec{V}_O - \vec{V}_m$

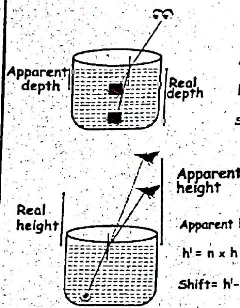
Velocity of image

$V_I = V_{I/m} + V_m$

Relative Velocity of image with respect to object

$V_{I/O} = V_{I/M} - V_{O/M}$

When object is in denser medium and observer in rarer medium



Apparent depth= $\dfrac{\text{real depth}}{n}$

$h' = \dfrac{h}{n}$

Shift= $h - h' = h - \dfrac{h}{n} = h\left(1 - \dfrac{1}{n}\right)$

Apparent height= n x Real height

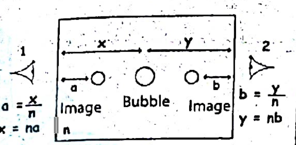
$h' = n \times h$

Shift= $h' - h = nh - h = h(n-1)$

Object is in denser medium → Shift is towards the surface

Object is in rarer medium → Shift is away from the surface

Air bubble in glass slab

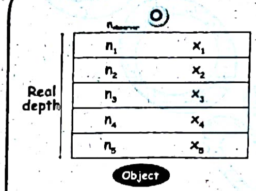


$a = \dfrac{x}{n}$, $x = na$

$b = \dfrac{y}{n}$, $y = nb$

Width of glass slab= $x + y = na + nb$

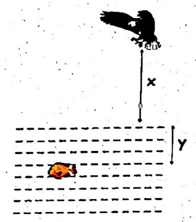
Multiple medium



Apparent height of object with respect to observer

$X = n_{observer} \sum \dfrac{x}{n}$

Bird fish problem

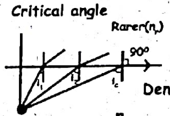


$V_{Fish/Bird} = n_{Bird}\left[\dfrac{x}{n_{Bird}} + \dfrac{y}{n_{Fish}}\right]$

$X_{Bird/Fish} = n_{Fish}\left[\dfrac{x}{n_{Bird}} + \dfrac{y}{n_{Fish}}\right]$

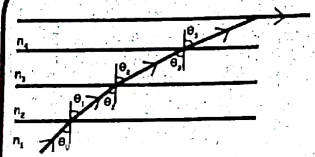
Total Internal Reflection

Critical angle



$i_c = \sin^{-1}\left(\dfrac{n_r}{n_d}\right)$

TIR in multiple medium



From snells law,

$n_1 \sin\theta_1 = n_2 \sin\theta_2 = n_3 \sin\theta_3 \ldots\ldots$

If $\sin\theta = 1$, means TIR Occured in a medium For TIR to occur, $i > i_c$

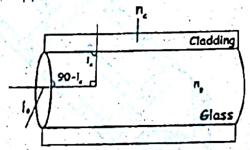
$\sin i > \sin i_c$

But, $\sin i_c = \dfrac{1}{\mu}$

$\therefore \sin i > \dfrac{1}{\mu}$

$\boxed{\mu > \dfrac{1}{\sin i}}$

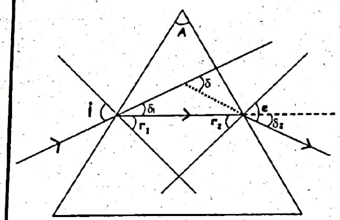
Applications of TIR Optical fiber cable



$\sin i_c = \dfrac{n_c}{n_g}$, $\sin i_o = \sqrt{n_g^2 - n_c^2}$

$\cos i_c = \dfrac{\sqrt{n_g^2 - n_c^2}}{n_g}$

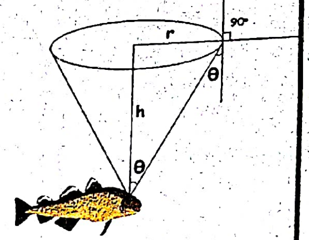
Prism



$r_1 + r_2 = A$

$\delta = i + e - (r_1 + r_2)$

Area of visible region
(From Under Water)



$$r = h \times \dfrac{1}{\sqrt{\left(\dfrac{n_d}{n_r}\right)^2 - 1}}$$

If $n_d = n$ and $n_r =$ (air) then,

$r = \dfrac{h}{\sqrt{n^2 - 1}}$

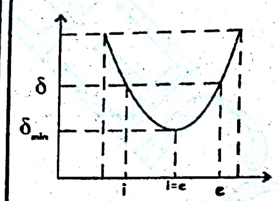
If $n_r = 1$; Area $= \dfrac{\pi h^2}{n^2 - 1}$

Angle of cone

Total angle $= 2 \times i_c$

$= 2\theta$

Deviation vs i graph



At δ_{min}, $i = \angle e$

Minimum Deviation

At minimum deviation:

1) $\angle i = \angle e$

2) $\angle r_1 = \angle r_2 = \angle r$

3) $\delta_{min} = D = i + e - (r_1 + r_2)$

$= i + e - A$

$D = 2i - A$

4) $2r = A$

5) Refractive index (n):-

$1 \times \sin i = n \times \sin r$

$n = \dfrac{\sin i}{\sin r}$

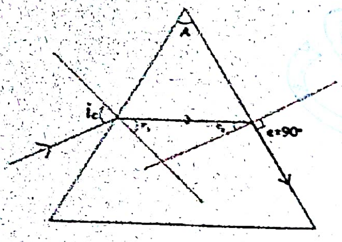
$$\boxed{n = \dfrac{\sin\left(\dfrac{A+D}{2}\right)}{\sin\dfrac{A}{2}}}$$

Note:-

If angle of prism = angle of minimum deviation i.e $\angle A = D$ then, $n = 2\cos(A/2)$

RAY OPTICS- 5

TIR in Prism



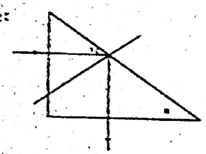
At second face,
$n \times \sin r_2 = 1 \times \sin 90$

$\sin r_2 = \frac{1}{n}$

$r_1 + r_2 = A$

$\sin i_c = n \times \sin r_1$

Note:



For TIR,
$i \geq i_c$
$n \geq \frac{1}{\sin i}$

Cauchy's Relation

$n = A + \frac{B}{\lambda^2}$

For VIBGYOR
$\lambda \rightarrow V < I < B < G < Y < O < R$
$n \rightarrow V > I > B > G > Y > O > R$

$\sin r_2 = \frac{1}{n} \Rightarrow i_c \propto \frac{1}{n}$

$i_c \rightarrow V < I < B < G < Y < O < R$

From V to R
λ:
n:
i_c:

Value of i for which rays will retrace its path

$\sin i = \left(\frac{n_2}{n_1}\right) \sin A$

Thin Prism

$\sin \theta \approx \theta$

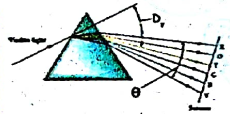
$$\boxed{n = \frac{\left(\frac{A+D}{2}\right)}{\frac{A}{2}}}$$

$D = (n-1) \times A$

Angular Dispersion (θ)

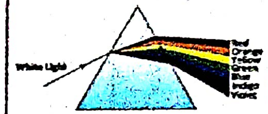
$\theta = (n^v - n^r) A$

Deviation of mean ray



$D_y = \left[\frac{n_R + n_V}{2} - 1\right] A$

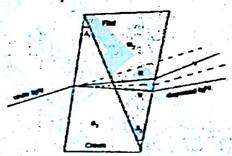
Dispersion in Prism



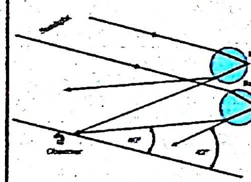
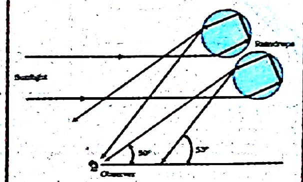
Cause:-
$n = A + \frac{B}{\lambda^2} \Rightarrow n \propto \frac{1}{\lambda}$
$D = (n-1)A \Rightarrow D \propto n$
Maximum deviation for violet
Minimum deviation for red
$D_{max} = (n_v - 1) A$
$D_{min} = (n_r - 1) A$
Mean ray → Yellow
$n_y = \frac{n_v + n_r}{2}$

Dispersive power

$\omega = \frac{n_v - n_R}{n_y - 1}$; $n_y = \frac{n_v + n_R}{2}$



$(n_1 - 1) A_1 = (n_2 - 1) A_2$

Primary Rainbow	Secondary Rainbow
	
One TIR Colours more clear Red colour on top and Violet on bottom 3 step process	Two TIRs Colours fainter Violet on top & red on bottom 4 step process

Some natural phenomenon due to Sunlight

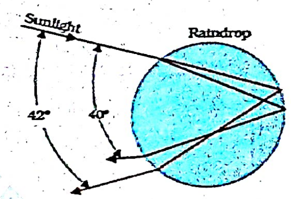
Rainbow

Combined effect of dispersion, refraction and reflection of sunlight by spherical water droplets of rain.

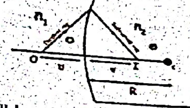
Condition for observing rainbow: Sun should be shining in one part and raining in opposite part of sky.
Observer can see rainbow only when his back is towards the sun.

Formation of Rainbow:

- Sunlight is refracted as it enters a raindrop Causing dispersion.
- Violet is bent most while red is bent least
- Light gets internally reflected if angle between refracted ray and normal to the drop surface is greater than critical angle (48°)
- Light is refracted again when it comes out of the drop
- Violet light emerges at an angle of 40° related to incoming sunlight and red light emerges at 42°



REFRACTION AT CURVED SURFACES



u = -ve
V = +ve
R = +ve

$$\frac{n_2}{v} - \frac{n_1}{-u} = \frac{n_2 - n_1}{R}$$

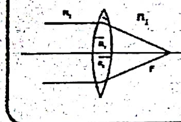
all lengths on the side of incident ray are taken as -ve
all lengths on the side of reflected ray are taken as +ve

$$\boxed{\frac{\text{I-O=S}}{\frac{\text{Medium}}{\text{I.Distance}} - \frac{\text{Medium}}{\text{O.Distance}} = \frac{\text{Change in medium}}{\text{Radius of curvature}}}}$$

Transverse Magnification $T.M = m = \frac{\frac{v}{n_2}}{\frac{u}{n_1}} = \frac{V \times n_1}{u \times n_2}$

LENS MAKERS FORMULA

To find focal length if refractive index is same on both sides of lens.

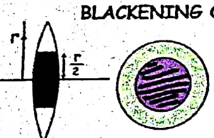


I-O=S $\quad \frac{n_1}{F} - 0 = \frac{n_2 - n_1}{R_1} + \frac{n_1 - n_2}{R_2}$

$$\boxed{\frac{1}{F} = \left[\frac{n_2}{n_1} - 1\right]\left[\frac{1}{R_1} - \frac{1}{R_2}\right]}$$

$$\boxed{\frac{1}{F} = \left[{}_1n_2 - 1\right]\left[\frac{1}{R_1} - \frac{1}{R_2}\right]}$$

BLACKENING OF LENS



Intensity ∝ Area of transmission of light
$I \propto A_T$

Blackening of lens $\Rightarrow A_T\downarrow \Rightarrow I\downarrow$

To find new intensity :
i) Find total A_T (Area of lens before blackening)
ii) Find new $A_T = (A_{total} - A_{opaque})$
iii) $I \propto$ original A_T (Total A_T)

$I' \propto$ new A_T

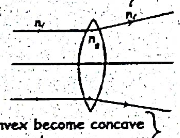
Taking ratio of these two equations we can find I'

Comparison of focal length in air & Liquid

1) If $n_{air} < n_{liquid} < n_{glass}$
- Nature of lens remains same
- Focal length increases

2) Same refractive index $[n_l = n_{glass}]$
- Lens become invisible

3) $n_l > n_g \quad f = \frac{R}{2\left(\frac{n_g}{n_l} - 1\right)}$ =negative



Convex become concave
Concave become convex } Nature of lens changes

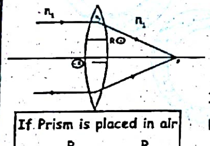
LENSES

Types
Thick lens - Two surfaces are at some distance apart.
Thin lens - Two surfaces are close

Thin Lens
- Convex: Biconvex, Equiconvex, Plano convex, Concavo convex
- Concave: Biconcave, Equiconcave, Plano concave, Convexo concave

Convex mirror, Convex lens } f=+ve
Concave mirror, Concave lens } f=-ve

EQUICONVEX LENS



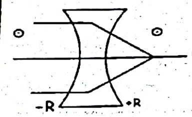
$$\boxed{F = \frac{R}{2[{}_1n_2 - 1]}}$$

If Prism is placed in air
$F = \frac{R}{2\left[\frac{n_2}{1} - 1\right]} = \frac{R}{2[n_2 - 1]}$

In water,
$F = \frac{R}{2\left[\frac{n_2}{n_w} - 1\right]}$

⇒ F↑ than air

FOCAL LENGTH OF EQUICONCAVE LENS IN AIR



$$\boxed{f = \frac{-R}{2[{}_1n_2 - 1]}}$$

PLANO CONVEX LENS

F of planoconvex lens = 2 × f of equiconvex lens

$$\boxed{f = \frac{R}{{}_1n_2 - 1}}$$

PLANO CONCAVE LENS

Equiconvex = f ⟶ Equi Concave = - f
⇓ ⇓
Plano Convex = 2f Plano Concave = -2 f

$$\boxed{f = -\frac{R}{{}_1n_2 - 1}}$$

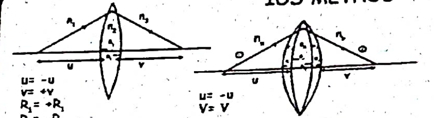
POWER

$$\boxed{\text{Power} = \frac{1}{F \text{ in metre}}}$$

[Power= 1 dioptre = $1m^{-1}$]

In centimetre, $\boxed{P = \frac{100}{f \text{ in cm}}}$

IOS METHOD



u= -u
v= +v
$R_1 = +R_1$
$R_2 = -R_2$
I-O=S

$$\frac{n_1}{v} - \frac{n_1}{-u} = \frac{n_2 - n_1}{R_1} + \frac{n_1 - n_2}{-R_2}$$

u= -u
V= V
$R = +R_1, +R_2, -R_3, -R_4$

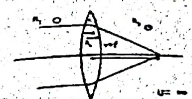
$$\frac{n_1}{v} - \frac{n_1}{-u} = \frac{n_2 - n_1}{R_1} + \frac{n_3 - n_2}{R_2} + \frac{n_2 - n_3}{-R_3} + \frac{n_1 - n_2}{-R_4}$$

u= -ve
V= V
$R_1 = -ve = -R_1$
$R_2 = \infty$

I-O=S

$$\frac{n_1}{v} - \frac{n_1}{-u} = \frac{n_2 - n_1}{-R_1}$$

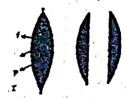
IOS METHOD FOR FOCAL LENGTH



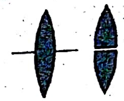
u = ∞
v = f
$R = +R_1, -R_2$

$$\frac{n_1}{f} = \frac{n_2 - n_1}{R_1} + \frac{n_1 - n_2}{-R_2}$$

CUTTING OF LENS

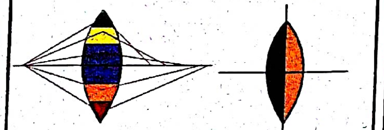


	Before cutting	After cutting
Focal Length	f	2f
Power	P	P/2
Area	A	A
Intensity	I	I



	Before cutting	After cutting
Focal Length	f	f
Power	P	P
Area	A	A/2
Intensity	I	I/2

NUMBER OF IMAGES FORMED IN MULTIPLE MEDIUM LENS



Number of medium =Number of images

Number of images = 1

LENS FORMULA

To find v when f and u are given

$$\boxed{\frac{1}{v} - \frac{1}{u} = \frac{1}{f}} \quad \frac{1}{v} = \frac{1}{f} + \frac{1}{u} = \frac{u+f}{uf} \quad \boxed{v = \frac{uf}{u+f}}$$

MAGNIFICATION

To find size of image

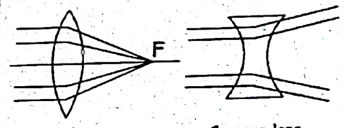
$$\boxed{m = \frac{[I]}{[O]} = \frac{v}{u}}$$

$$\boxed{m = \frac{f}{f+u} \qquad m = \frac{f-v}{f}}$$

m_T:
- $|m_T| = 1 \Rightarrow$ Same size
- $|m_T| < 1 \Rightarrow$ Small
- $|m_T| > 1 \Rightarrow$ magnified

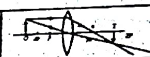
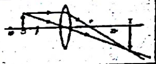
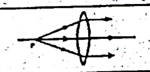
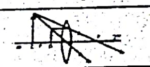
m_T:
- +ve ⇒ Virtual image, Erect
- -ve ⇒ Real image, Inverted

IMAGE FORMATION BY CONVEX AND CONCAVE LENSES

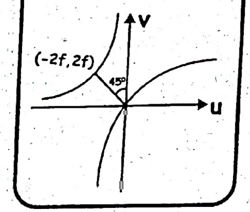


Convex lens
- Converging lens
- Similar to concave mirror

Concave lens
- Diverging lens
- Similar to convex mirror

Sl NO	Position of the object	Ray Diagram	Position of image	Nature of image
1	At $\lvert u\rvert = \infty$		At F	Real, inverted, diminished
2	Beyond 2F $\lvert u\rvert > 2\lvert f\rvert$		Between F and 2F	Real, inverted, diminished
3	At 2F $\lvert u\rvert = 2\lvert f\rvert$		At 2F	Real, inverted, same size
4	Between F & 2F $f < \lvert u\rvert < 2f$		Beyond 2F	Real, inverted, enlarged
5	At F $\lvert u\rvert = \lvert f\rvert$		At ∞	Cannot be defined
6	within F $\lvert u\rvert < \lvert f\rvert$		On the side of object	Virtual, erect, enlarged

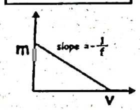
U - V GRAPH



MAGNIFICATION VS V GRAPH

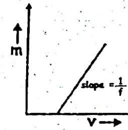
Concave lens

$$\boxed{m = \frac{f-v}{f} \qquad m = 1 - \frac{V}{f} = -\frac{V}{f} + 1}$$

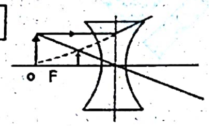


Convex lens

$$\boxed{m = \frac{V}{f} + 1}$$

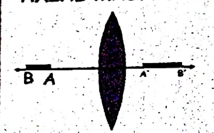


Concave lens



-Real object
-Virtual image
-Formed between F and Pole

AXIAL MAGNIFICATION

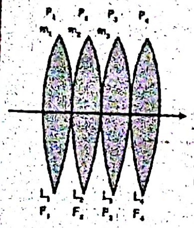


$m_L = \frac{\text{length of image}}{\text{length of object}} = \frac{A'B'}{AB} = \frac{V_2 - V_1}{U_2 - U_1}$

For short object
$m_L = \frac{dv}{du} = m_T^2$

COMBINATION OF LENSES

Lenses are combined such that there is no gap between them



Power

$P_{comb} = P_1 + P_2 + P_3 + ...$

Focal length

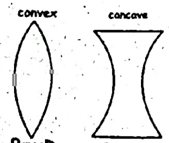
$\frac{1}{f_{comb}} = \frac{1}{f_1} + \frac{1}{f_2} + \frac{1}{f_3} + ...$

Magnification ,

$m_T = m_1 \times m_2 \times m_3$ .....

Note :

P_{comb} determines whether the combination act as converging lens or diverging lens.



P=xD P=-yD

System acts as converging lens if total power > 0

$P_T > 0$

$\Rightarrow P_{convex} > P_{concave}$

$\Rightarrow f_{concave} > f_{convex}$

System acts as diverging lens if total power < 0

$P_T < 0$

$\Rightarrow P_{concave} > P_{convex}$

$\Rightarrow f_{convex} > f_{concave}$

System acts as plane lens /glass if $P_{comb} = 0$

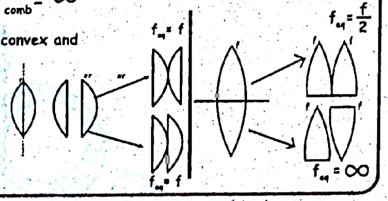
$P_{concave} = P_{convex} = 0$

$f_{concave} = f_{convex} = f_{comb} = \infty$

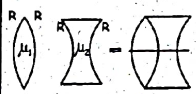
For a combination of convex and concave lenses

If $P_{convex} > P_{concave}$

→ combination acts as convex lens



F_{EQ} USING LENS MAKERS FORMULA



$$f_{eq} = \frac{R}{2\,[\mu_1 - \mu_2]}$$

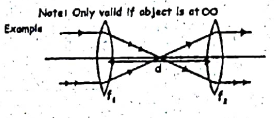
LENSES COMBINED SUCH THAT THERE IS A GAP BETWEEN THEM

Power : $P_{comb} = P_1 + P_2 - dP_1P_2$

$$\frac{1}{f_{comb}} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1 f_2}$$

Note : Only valid if object is at ∞

Example



If $P_{comb} = 0$

$$\frac{1}{f_{comb}} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1 f_2} = 0$$

$$\boxed{d = f_1 + f_2}$$

NATURE OF MIRROR IS DETERMINED BY FOCAL LENGTH

To find focal length, split the silvered lens into a lens and a mirror and apply IOS

$u = \infty$ I - O = S

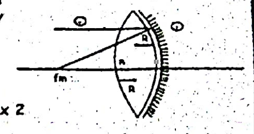
$v = f_m$

$R_1 = +R$ $\frac{-1}{f_m} = \frac{n-1}{R} \times 2 + \left[\frac{0+n}{+R}\right] \times 2$

$R_2 = -R$

[Ray passes through the surface 2 times thus multiplied by 2]

$$f_m = \frac{-R}{4n-2} = \frac{-R}{2(2n-1)}$$



When equiconvex lens is silvered (mirrored) its focal length become negative (concave mirror) with magnitude

$$f_m = \frac{-R}{2(2n-1)}$$

OPTICAL INSTRUMENTS

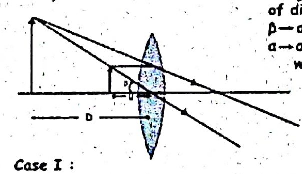
Simple Microscope

Only one lens [convex lens]

Image is formed at least distance of distinct vision (D)

β→angle subtended by image at eye.

α→angle subtended by object at eye when placed at distance D.



Case I :

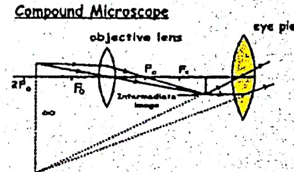
Eye under relaxed state or normal vision Final image at infinity

Object at f→ $u = u_{max}$ $\boxed{m_{min} = \frac{D}{u_{max}} = \frac{D}{f}}$

Case 2 : Eye under strain

Final image at D $u = u_{min}$ $\boxed{m_{max} = \frac{D}{u_{min}} = 1 + \frac{D}{f}}$

Compound Microscope



-Has 2 lenses

- Length of microscope

= Distance between lenses.

-Magnification $m = m_o \times m_e$

[m_e - same as simple microscope]

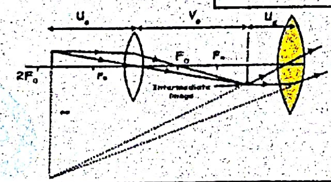
To get magnified image, object placed between F_0 & $2F_0$ of objective lens. This image is called intermediate image.

Intermediate image→real, inverted , magnified.

Intermediate image is formed within or at the focus of eyepiece.

Total magnification $m_T = -m_o \times m_e$ $\boxed{m_T = -m_o \times m_e = -\left[\frac{V_o}{u_o} \times \frac{D}{u_e}\right]}$

m_e same as simple microscope



SILVERING OF PLANO-CONVEX LENS

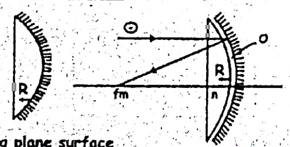
2 possibilities

i) silvering curved surface

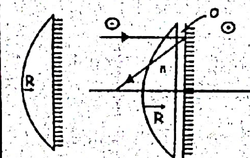
ii) Silvering plane surface

Silvering curved surface

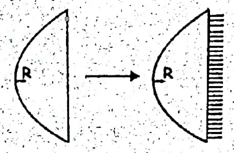
$$f_m = \frac{-R}{2n}$$



Silvering plane surface



$$f_m = \frac{-R}{2(2n-1)}$$

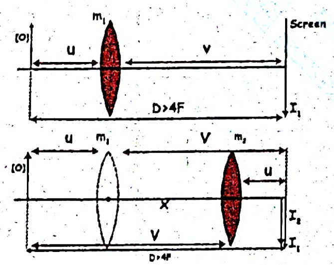


Before silvering After silvering

$$f = \frac{R}{(n-1)} \qquad f_m = \frac{-R}{2(n-1)} = \frac{f}{2}$$

LENS DISPLACEMENT METHOD

Distance between object & image > 4F



In lens Displacement method

1) D ≥ 4F

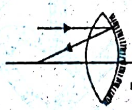
2) $F = \frac{D^2 - x^2}{4D}$ x→distance between 2 position of lens

3) $F = \frac{x}{m_1 - m_2}$ F= Focal length of lens

4) $m_1 m_2 = 1$

$[O] = \sqrt{I_1 I_2}$

SILVERING OF LENS



Lens → Mirror

$$m_T = \frac{v}{u} = \frac{[I]}{[o]} = \frac{f}{f-u} = \frac{f-v}{f}$$

LENS MIRROR COMBINATION

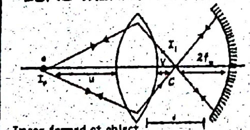
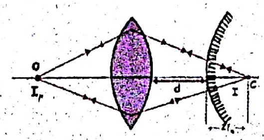


Image formed at object

If both lens and mirror are converging. $d = v + 2f_m$



If one is converging & other is diverging $d = v - 2f_m$

Case I :

Eye in relaxed state or final image at ∞, $u_e=f_e$

$m_T=m_{min} = -\left[\frac{V_o}{u_o}\ \frac{D}{f_e}\right]$ Also $L=L_{max}=V_o+u_{max}=V_o+f_e$

Case 2 : Strained eye

$$L=L_{min}=L_D=V_o+\underset{u=u_{min}}{U_e}=V_o+u_{min}=V_o+\frac{Df_e}{D+f_e}$$

$\boxed{m_D=m_{max}=\frac{-v_o}{u_o}\left[1+\frac{D}{f_e}\right]}$ $\quad \frac{m_{max}}{m_{min}}=\frac{D+f_e}{D}$

$$\boxed{\begin{aligned} m_D&=\left[\frac{L}{f_o}\right]\left[1+\frac{D}{f_e}\right]\\ m_\infty&=\left[\frac{L}{f_o}\ \frac{D}{f_e}\right]\end{aligned}}$$

Note :

$L_\infty=V_o+U_{max}$ $\quad L_\infty=V_o+f_e$

$L_\infty=\frac{U_o f_o}{U_o-f_o}+f_e$ $\quad L_D=V_o+U_{min}$

$L_D=V_o+\frac{Df_e}{D+f_e}$ and $L_D=\frac{U\ f_o}{U_o-f_o}+\frac{Df_e}{D+f_e}$

For microscope, eyepiece larger than objective

$f_e\uparrow \Rightarrow m\uparrow \quad f_e\downarrow \Rightarrow m\downarrow$

For telescope, eyepiece smaller than objective to increase magnification.

Telescope

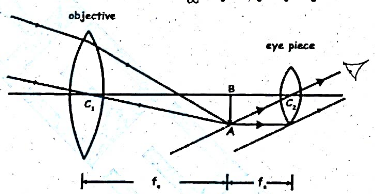
Magnification $m=\frac{f_o}{u_e}$ $\quad u=u_{max} \Rightarrow m=m_{min}$; $\quad u=u_{min} \Rightarrow m=m_{max}$

Length of telescope $L=f_o+u_e$

Normal adjustment /Relaxed eye/final image at ∞

$m_{min}=\frac{f_o}{f_e}$ $\quad$ Length $=L_\infty=f_o+u_e=f_o+f_e$

$\boxed{L_\infty=f_o+f_e}$



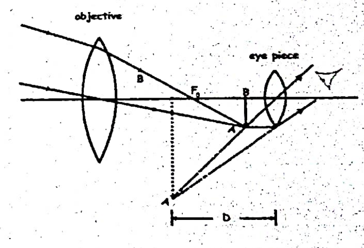
Eye under strain/Final image at least distance of distinct vision.

$u=u_{min}=\frac{Df_e}{D+f_e}$

$m_{max}=\frac{f_o}{f_e}\left[1+\frac{f_e}{D}\right]$

$L_D=f_o+u_e$

$L_D=f_o+\frac{Df_e}{D+f_e}$



Length of Telescope

u_e can have values : f_e or $\frac{Df_e}{D+f_e}$

v_o can have values : f_o only

$L_D=v_o+u_e$